

Canadian Short-term Rentals and their Impact on Long-term Rental Prices

A Review and Critique of the Canadian Literature

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Executive Summary

The STR market has been growing and accelerating, facilitated by platforms like Airbnb and Vrbo. This has raised questions as to whether and to what magnitude are STRs contributing to the housing affordability problems and by how much have STRs increased long-term rental prices? The evidence is sparse and the results that do exist conflict: some research suggests that there is no evidence that STRs impact long-term rental prices and others suggest that STRs increase long-term rental prices by a large magnitude. In this report, we critically examine the Canadian research to understand why these conflicting results are found. Along with examining the conclusions reached by the papers/reports, we examine, compare, and critique the data and methodology used in answering the research question.

We examine three high-profile reports on STRs in Canada. First, Dr. David Wachsmuth released a report on STRs in British Columbia and found that “an increase of one dedicated STR per 100 rental units predicts that average rents in the neighbourhood will be \$49 higher”. The Conference Board of Canada released their report on Canadian STRs and found “no compelling evidence that the level of Airbnb activity had a meaningful impact on rents” (Conference Board of Canada 2023). Desjardins released a brief that concluded “research has established that there is a relationship between the prevalence of STRs in a community and higher rents for LTRs” (Desjardins 2023).

How is it that three reports on Canada’s STRs reach differing conclusions? Differing conclusions were reached because the reports asked different questions and used different methodologies to answer their respective questions. However, due to the methodologies and interpretations of results used in the reports, the findings should be approached with caution. While these reports do inform us about some aspects of the STR market in Canada, but they do not sufficiently answer the question of whether an increase of STRs in Canada *cause* long-term rental prices to increase, or vice versa, whether restrictions of STRs in Canada would cause long-term rental prices to permanently decrease. They also do not answer by how much STRs may cause long-term rental prices to change. The research in this area is very nascent and more careful research is required, particularly research that is focused on providing causal evidence.

While these reports may not satisfactorily answer the question of whether STRs cause long-term rental prices to increase (or, conversely, whether banning STRs would cause long-term rental prices to decrease), these reports are nevertheless welcome contributions to this important policy question. The difficulty with the empirical models that we highlight here is not surprising as it continues to be a hurdle to overcome in other jurisdictions: other jurisdictions report equally conflicting results and encounter similar methodological challenges. This will likely remain an important policy question and methodological challenge for years to come.

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1. Introduction

It is not unusual to find strong opinions when it comes to the question of what should be done with short-term rentals (STRs). Should they be regulated? If so, how strictly should they be regulated? Or should they just be banned? And one's opinion on how the market should be regulated is likely directly related to one's belief on how STRs affect rental and housing availability and affordability. Canada is currently facing a housing crisis: housing costs have been on the rise and in the long-term rental market, vacancy rates are very low and rental prices are very high (CMHC 2023). At the same time, the STR market has been growing and accelerating, facilitated by platforms like Airbnb and Vrbo. The question is whether and to what magnitude are STRs contributing to the housing affordability problems? Of particular interest is whether and by how much have STRs increased long-term rental prices? While there are a handful of studies from the US that empirically examine this question, the evidence is even sparser in Canada. Furthermore, the empirical evidence that does exist in Canada finds conflicting results with some research suggesting that there is no evidence that STRs impact long-term rental prices and others suggesting that STRs increase long-term rental prices by a large magnitude. In this note, we critically examine this Canadian research to understand why these conflicting results are found. Along with examining the conclusions reached by the papers/reports, we examine, compare, and critique the data and methodology used in answering the research question.

Recently, three high-profile reports on STRs in Canada have been released. Beginning in August 2023, Dr. David Wachsmuth released a report on STRs in British Columbia and found that “an increase of one dedicated STR per 100 rental units predicts that average rents in the neighbourhood will be \$49 higher” (Wachsmuth 2023). In October 2023, The Conference Board of Canada released their report on Canadian STRs and found “no compelling evidence that the level of Airbnb activity had a meaningful impact on rents” (Conference Board of Canada 2023). Finally, in December 2023, Desjardins released a brief that concluded “research has established that there is a relationship between the prevalence of STRs in a community and higher rents for LTRs” (Desjardins 2023).¹

How is it that three reports on Canada's STRs reach differing conclusions? We argue that these differing conclusions were reached because the reports asked different questions and used different methodologies to answer their respective questions. Despite the different questions and methodologies, we also argue that, due to the methodologies and interpretations of results used in the reports, the findings should be approached with caution.² Overall, these reports do inform us about some aspects of the STR market in Canada, but they do not sufficiently answer the question of whether an increase of STRs in Canada *cause* long-term rental prices to increase, or vice versa, whether

¹ There is a fourth paper this is sometimes mentioned. In 2020, the Canada Mortgage and Housing Corporation (CMHC) released an evidence brief on the impact of short-term rentals on Canadian housing (CMHC 2020). While the detailed report upon which this evidence brief was not published by CMHC is was later published in a peer reviewed journal (Combs, Kerrigan, and Wachsmuth 2020). The paper did not consider the effect of STRs on rents or housing affordability—instead focusing solely on short-term rental activity—and, therefore, is out scope for this brief.

² Note that even in the US literature that examines the causal impact of STRs on long-term rental prices, methodology is also suspect: there has been no definitive methodology or paper that adequately answers the causal question of the impact of STRs on rental prices.

restrictions of STRs in Canada would cause long-term rental prices to permanently decrease. They also do not answer by how much STRs may cause long-term rental prices to change. The research in this area is very nascent and more careful research is required, particularly research that is focused on providing causal evidence.

This note begins with a high-level review of important concepts. This is followed with a comparison of the data and methodology used in the reports. Finally, we discuss and conclude.

2. Important Concepts

Before delving into the papers mentioned above, it is important to distinguish between association and causation. Readers may be familiar with the saying “correlation is not causation.” Correlation implies association—two things move together—but that does not imply that one of those things *causes* the other thing to move. That is, correlation or an association means that there is a statistical relationship between two things. For example, there may be an association between the sale of air conditioners and the sale of ice cream: as more ice cream is sold, more air conditioning units are sold and vice versa. On the other hand, causality asks whether one event *causes* the other: does an increase in the sale of ice cream *cause* an increase in the sale of air conditioners? In this example, there is likely no causal relationship. An increase in the sale of ice cream and an increase in the sale of air conditioners are more likely caused by an increase in temperature: there is a mediating or *confounding* factor. For there to be causation there must be both association *and* effect. Effect, that A causes B, is much more difficult to prove than association. Proving causation requires that, in addition to an association, that the cause came *before* the effect, there is no confounding factor, there is a causal mechanism, and there is a causal context.

The same principle must be kept in mind when asking: what is the effect of STRs on long-term rental prices? On the one hand there may be an association: more STRs may be *associated* with higher long-term rental prices. This may occur because the most profitable locations for STRs are in city centres or high-tourist areas while at the same time the most desirable locations for long-term rentals are in city centres and high-tourist areas (e.g., downtown Calgary, Canmore, etc.). This association is different from causation. Causation asks whether an increase in STRs *causes* an increase in long-term rental prices. One potential reason we may think STRs cause an increase in long-term rental prices (and the hypothesis we may want to test) is: if STRs remove housing from the long-term rental market, this can decrease long-term rental supply and thus increase long-term rental prices.

Following from that point, the second important concept to keep an eye on is *omitted variable bias*. When testing for causation, we must be sure to include in our model all variables that may simultaneously affect the number of STRs (the “dependent” variable) and long-term rental prices (the “explanatory” variable). This would include things like population growth, number of STR and LTR units, economic factors, and related factors. If we do not include all potential variables that may affect both,

this can ‘bias’ the measurement of the effect of STRs on long-term rental prices.³ That is, the magnitude of the effect will be mismeasured.

One way of combatting omitted variable bias when the omitted variable is not observed is using a difference-in-difference model (DID). In a simple DID model, there are two groups and two time periods. One group is “treated” and the other group is not treated, e.g., the “control”. The treated group is compared to the control group both before and after the treatment. This allows us to ask whether the treatment caused the treated group to experience an outcome different from the control group. The logic is that had the treated group not been treated, they would have had the same post-treatment outcome as the control group. In looking at the case of STRs, the question might be whether the implementation of STR regulations cause long-term rental prices to decrease. In this example, we would compare a municipality with STR regulations and a municipality without STR regulations before and after the implementation of STR regulations. For a DID model to be causal, it is imperative that the control group looks like the treatment group in every respect except for the treatment. That is, at minimum, the municipality with STR regulations must have had long-term rental prices that were moving the same way—increasing or decreasing at the same rate over time—as the municipality without regulations before the implementation of regulations. That is, the “parallel trends” assumption must be met. This assumption must hold to make the causal claim that, had a municipality that did implement STR regulations not implemented them, they would have had the same long-term rental prices that the municipality with no regulations has. This is at minimum shown graphically, although more technical regression methods have been developed to check this assumption.

A second important feature to take account of in DID is “staggered timing”. In simple DID models as discussed above, there are only two groups (one treatment and one control) and two time periods (before and after the treatment). In reality, DID models are not this simple: there are often more than two groups and more than two time periods. In the STR example, we might compare all large municipalities in Canada from 2015 to 2023. Some of these municipalities will have STR regulations, and those regulations will have been implemented at different times over the years (e.g., staggered timing”). In this case, a careful (i.e., more complex) empirical specification must be specified to ensure that when comparing municipalities with regulations to municipalities with no regulations post-regulation implementation, that only municipalities with no regulations *at the time of comparison* are used as the comparator. If municipalities with regulations are compared to both municipalities with regulations and municipalities without regulations (e.g., the empirical specification does not take into account staggered timing), the magnitude of the coefficient of interest will be biased (i.e., incorrect): it creates “bad” treatment combinations, biasing the estimation, to the point of reversing the sign.⁴

³ The bias being referred to here is called statistical bias. When you leave out an important in an analysis it means that its effect can instead be included in the measurement of the effect of the other variables that are included in your analysis. This means that the effect of the other variables included in the analysis can be overestimated or underestimated and, as such, are statistically biased.

⁴ See: Goodman-Bacon (2021) and Roth et al. (2023) for more detailed explanations.

3. A Comparison of the Reports

3.1 Research Questions and Interpretation of Results

As discussed in the introduction, three reports on the effect of Canadian STRs on long-term rental prices have been released recently: the Wachsmuth report, the Conference Board Report, and the Desjardins report. Each report asked a different research question—which, although seemingly only slightly different—has important implications for the methodology used and the interpretation of the conclusions.

The **Wachsmuth report** examined the *association* (i.e. correlation) between STRs and long-term rental prices in British Columbia and finds that a city with more STRs is associated with higher long-term rental prices. This effect is not necessarily *caused* by STRs: it could be that STRs and long-term rentals face similar levels of demand in a neighborhood, affecting both the supply of STRs and the price of long-term rentals.

Comparatively, the **Conference Board report** examines the *causal* impact of STRs on long-term rental prices in Canada. They find no evidence of a causal impact, meaning that they find no evidence that STRs *cause* an increase in long-term rental prices but that they may still be associated.

Finally, the **Desjardins brief looks** at whether restricting STRs could alleviate Canada’s housing crisis. While they were not explicitly clear on their research question (e.g., whether they were examining association or causation), based on their literature review, they conclude that “research has established that there is a relationship between the prevalence of STRs in a community and higher rents for LTRs” (Desjardins 2023)—suggesting they examined associations. However, their literature review includes literature from the US and Europe (which is not necessarily applicable in Canada) and does not distinguish between literature that examines causal impacts and literature that examines associations.

3.2 Data and Definitions

In this section, we compare the data and definitions the reports use. We focus on the Wachsmuth report and the Conference Board Report as both use an empirical approach to answer their research questions. The Desjardins report is a literature review and a presentation of descriptive data with no empirical model.

To begin, the reports use different data sources. The Wachsmuth report uses Airbnb and VRBO listings scraped by AirDNA. AirDNA scrapes STR listings daily and, using machine learning, calculates occupancy rates and revenue. While this data does not come direct from Airbnb and is based on a machine learning algorithm as opposed to the “actual” data, it is the next best data source and is often used in academic literature (Airbnb very reluctantly shares their data). The Conference Board of Canada uses data directly from Airbnb (their report was commissioned by Airbnb). Regardless of data source, it is unclear to what extent the data has been cleaned in either report. For example, traditional tourist accommodation providers like hotels often list rooms on Airbnb. Whether either report removes such listings from their

data is unclear. Finally, both reports use the CMHC rental survey, conducted in October of every year, for their long-term rental prices. In the absence of better data, this is the best data currently available on long-term rental prices in Canada.⁵

Both reports focus on the effect of “dedicated” STRs on long-term rental prices. “Dedicated” STRs are defined differently. The Wachsmuth report defines a dedicated STR as an STR that has been actively listed on a site like Airbnb or VRBO for at least 183 out of the last 365 days (e.g., six months) and booked for at least 90 out of the last 365 days (e.g., three months). For STRs that have not been around for at least a year (e.g., for new listings), a statistical model is applied using three months of listing patterns to predict whether an STR listing will be a dedicated listing. In the Conference Board Report, a dedicated STR is an STR that is booked for at least 30 days of the last three months.⁶

It could be argued that both definitions of dedicated STRs are problematic for two reasons. First, they are subjective. There is no theoretical grounding upon which to base their choice. Second, and more problematically, both only consider one aspect of duration: the total number of days in a given time span. However, when examining durations, number of “spells” and length of “spells” should be considered in combination with total number of days. A spell is a continuous amount of time a STR listing is listed on a platform for. Some spells may occur once and be short, e.g., an STR operator who lists their house for one week during an event like Stampede. A spell may also occur multiple times for short periods, e.g., an oil sands worker who lists their Calgary house as an STR every two weeks when they are away at work but lives there otherwise. A spell may also occur infrequently but be of mid-duration, e.g., a couple who take a temporary work position in Mexico for four months but otherwise lives in Calgary. Finally, spells may occur infrequently but be of long duration, e.g., a property that is listed as an STR for two years continuously while the owner is working in Houston who then returns to Calgary to live in the house permanently (perhaps with small breaks for short returns for upkeep and maintenance).⁷

While the definition of “dedicated” does not directly impact what the reports can teach us, it is a cause for concern. There is no standardized method for determining what a “dedicated” STR is. When interpreting results, the definition of dedicated STR must be kept in mind: if the definitions are different, the effects of different types of STRs are being measured.

3.3 Empirical Models

The empirical model in the Wachsmuth report is a predictive linear regression model. The average long-term rental price of a neighborhood is regressed on the number of dedicated STRs per 100 private dwellings, a time trend, the percent of dwellings that are renter-occupied, and dichotomous variables for type of municipality/area (e.g., mid-sized cities, core cities, large regions, non-urban, & resort

⁵ Statistics Canada has recently entered a partnership with rentals.ca to launch a new rental housing market index that will provide much more robust information about Canada’s rental housing market.

⁶ We disagree with these approaches to defining a dedicated STR both of which are subjective. In our work in Calgary, we use an innovative approach that uses a cluster analysis that removes the arbitrariness of setting a ‘dedicated’ threshold and which considers different dimensions of duration, using both spells and days in a spell.

⁷ For more information on why this matters and what can be done to address it, see Petit and Tedds (2023).

towns). This empirical specification is not causal and there is a problem of omitted variable bias that is not accounted for. Namely, if the most profitable locations for STRs are in the same areas as for long-term rentals, neighborhood effects should, at minimum, be controlled for. While municipality type is controlled for, given the very location-specific nature of STRs, it is unlikely that the type captures the granularity accurately. This is not an issue given that the model was not intended to be causal; however, it should be kept in mind when interpreting the results.

A more pressing issue with the empirical model in the Wachsmuth report is the interpretation of the coefficient related to the impact of STRs on rental prices. The report finds a coefficient of 48.65 on the number of dedicated STRs per 100 private dwelling and interprets the coefficient as meaning “an increase of one dedicated STR per 100 rental units predicts that average rents in the neighborhood will be \$49 higher” (p. 17). This is not the correct interpretation. This should more properly say that an increase of one dedicated STR per 100 rental units predicts that average rents in the neighborhood will be \$0.49 higher or an increase of 100 dedicated STRs per 100 rental units predicts that average rents in the neighborhood will be \$49 higher.⁸

The first empirical model in the Conference Board report is a fixed effects model. In the fixed effects model, the log of long-term rents in a neighborhood at time t are regressed on the number of dedicated STRs divided by the number of dwellings in a neighborhood at time t , a neighborhood fixed effect and a time-municipality fixed effect.⁹ In this model, they find that “a one percentage point increase in the share of the number of high-use Airbnbs to the number of households in a neighborhood is associated with, on average, an increase in rent by 0.2 percent. This result is not statistically significant” (p. 17). It is clear that, and the Conference Board confirms, this fixed effects model examines association, not causation. Regardless, this model partially addresses the omitted variable bias as discussed above: it includes a neighborhood fixed effect to absorb any unobservable, permanent difference in neighborhoods that simultaneously affect the decision to operate an STR in that neighborhood and charge higher long-term rental prices. However, unobservable time-varying neighborhood factors have not been accounted for. For example, changing neighborhood amenities (e.g., “urban revival”) may make neighborhoods simultaneously more amenable to STRs and long-term rentals (or vice versa). This creates a reverse causality problem: as desirable neighborhood amenities increase, the demand for long-term rentals in a neighborhood may increase, increasing long-term rental prices, and allowing STR operators to increase their STR prices in the same neighborhood (spillover effects), thus increasing the number of STRs in the neighborhood. The Conference Board Report does not control for these time-

⁸ The estimate is as follows: the predicated long-term average rental price in a community = $48.65 * \frac{1 \text{ Dedicated STR}}{100 \text{ Rental Units}}$. The proper interpretation of this coefficient is that a one unit increase in the *ratio*, that is 100 dedicated STRs per 100 rental unit, leads to a change of \$48.65 in the rental price. If we want to know the impact of 1 dedicated STR on the long-term average long-term rental price then we want to know the impact of a $\frac{1}{100}$ unit change in the ratio, which is 0.4865, or round to 0.49.

⁹ Note that a number of models in this vein are run (“robustness checks”) including lagging the share of dedicated STRs by one and two years and regressing the log of long-term rental rents on the number of booked nights. The results all reach the same general conclusion.

varying unobserved effects, making the results from this model an association and not causation, as well as biased.¹⁰

The second model in the Conference Board Report is a difference-in-difference (DID) model. This is the model that the Conference Board claims support the argument that STRs have no *causal* effect on long-term rental prices. The purpose of the DID model is to examine whether the introduction of personal residence regulations (PRR) had a *causal* effect on long-term rental prices. In this model, they compare municipalities that implement PRR's (the "treatment") and municipalities that do not implement PRRs (the "control"). The hypothesis is that if STRs have a positive causal effect on long-term rental prices, the introduction of PRR's should reduce long-term rental prices, and this effect should be larger for neighborhoods with a higher share of dedicated STRs.

In the Conference Board DID model, the log of average long-term rental prices is regressed on the share of dedicated STRs in a neighborhood at time *t*, a dichotomous treatment variable that equals one if there is a PRR effective in the neighborhood at time *t*, and an interaction of the share of dedicated STRs and the PRR dummy. The findings from this model are "despite having a clear effect on STR activity, these regulations did not lead to lower rents in the jurisdictions in which they were implemented" (p. 13). That is, the estimated coefficient on the interaction term was 0.0115 and it was not statistically significant suggesting that restricting STRs to personal residences did not decrease long-term rental prices as would have been expected if STRs in fact caused long-term rental prices to increase.

The DID specification in the Conference Board Report is incomplete. It is not clear if the parallel trends assumption is satisfied —no evidence either way is provided though the assumption is noted (p. 6 & p. 10)— nor does it appear to take into consideration the staggered timing of personal residence regulations that have occurred across Canada.¹¹ Evidence related to both the parallel trends assumption and staggered timing is required to make the argument that what is being measured is a causal effect and that the coefficient is not biased. Thus, the DID model and the resulting estimates in the Conference Board Report should be treated with caution: they may not be causal (without further evidence, there is no evidence for or against causality).

4 Discussion and Conclusion

Overall, although the three reports reviewed all examine the effect of STRs on long-term rental prices, they reach different conclusions. On the one hand, they answer different questions. The Wachsmuth report examines the *association* between the share of dedicated STRs (e.g., STRs that are listed for at least six months in a one-year period) and long-term rental prices. The Conference Board report claims to examine the *causal* impact of dedicated STRs (e.g., STRs that are booked at least 30 days over 3 months) on long-term rental prices. The Desjardins report appears to examine the *association* between STRs and long-term rental prices; however, their exact question is unclear.

¹⁰ Much of the academic literature on STRs is centered on resolving this identification issue. See for examples: Barron, Kung, and Proserpio (2021) and García López et al. (2020).

¹¹ No mention of addressing staggered timing is made in the publicly available report.

The conclusions of the reports should be approached with caution. The Wachsmuth report appears to overstate the strength of the association between STRs and long-term rental prices. The Conference Board report requires more research (e.g., correcting omitted variable bias, and addressing the parallel trends assumption and staggered timing) to conclusively state that STRs have a causal effect on long-term rental prices. Finally, the Desjardins report should untangle associations and causality, and, arguably, when providing recommendations for Canada, should focus on Canadian evidence.

None of the reports provide conclusive evidence to make the case that if STRs were restricted in Canada (or BC), that long-term rental prices would decline. Both the Wachsmuth report and Conference Board report provide evidence for a small positive association; however, the Conference Board report suggests that this association is not statistically significant. While this suggests the possibility that long-term rental prices may decrease slightly if STRs were to be banned in the short-run, it is unclear if mid- and long-term rental prices would decline. That is, if STRs were restricted some STRs could return to the long-term rental market, likely decreasing long-term rental prices in the short run. However, as the rental market adjusts to the “new normal”, it is uncertain whether this effect will last into the mid- and long- run without further causal evidence. While the Conference Board report is a start at modelling the casual impacts of STRs on long-term rental prices in Canada, it remains incomplete, and more research is required to state that this result would hold conclusively.

While these reports may not satisfactorily answer the question of whether STRs cause long-term rental prices to increase (or, conversely, whether banning STRs would cause long-term rental prices to decrease), these reports are nevertheless welcome contributions to this important policy question. The difficulty with the empirical models that we highlight here is not surprising as it continues to be a hurdle to overcome in other jurisdictions: other jurisdictions report equally conflicting results and encounter methodological challenges. This will likely remain an important policy question and methodological challenge for years to come.

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